THE LEGEND OF POS

(Poverty of Stimulus)

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Abstract

The Poverty-of-Stimulus (POS) argument is central to *Universal Grammar* (UG), and UG is central to Generative Grammar. Without UG, the abstract grammatical derivations postulated in Generative grammars are just formal exercises, as no one can reasonably expect children to successfully learn the intricate rules of such brain teasers.

There is a growing awareness that the claims regarding PoS and UG are unfounded. The arguments in favour of PoS do not live up to their originator's rhetoric promises, and moreover, as will be argued, UG is an assumption that is not needed for an empirically adequate grammar theory. So, it is time to bid farewell to PoS and to UG, for there is a superior account

1. Background

UG is to Generative Grammar what the *luminiferous aether* is to 19th century physics. Without the respective assumptions, the theories would not work. Michelson & Morley's (1887) experiments put an end to the aether theory. The UG theory has come to its end, too, but not because of brilliant experiments, but because it is an empirically unfounded assumption that is no longer needed, even if orthodox Generativists don't want to know anything about this.

The *Poverty-of-Stimulus* (POS) argument is foundational to the UG theory, serving as the primary 'evidence' that humans 'must' possess a *genetically endowed* language faculty. The linguistic input that children receive ('stimuli') is allegedly too limited and underdetermined to explain the rapid acquisition of a complex grammar by a ±four-year-old in any other way. PoS posits that without innate cognitive channelling or priming by UG, acquiring grammatical rules from limited and partially variant data would not be possible. PoS has thus become a key testimony to the indispensability of UG.

However, the "*Poverty of Stimulus*" argument is also self-defeating as it simultaneously serves as a carte blanche for postulating arbitrary features of presumed grammars. Since UG is seen as the code book for deciphering grammars, it serves as a general permission to spread even extremely abstract ideas without restraint that are so far removed from the data that hardly anyone who independently examines the data would ever arrive at the same analysis, but *every child* is allegedly able to easily crack them thanks to a genetically encoded, innate UG. PoS legitimates the assumption of UG and on the other hand, UG is the indispensable background assumption for a Generative grammar of a language: "*The theory of the genetically based language faculty is called Universal Grammar; the theory of each individual language is called its Generative Grammar*." (Chomsky 2017: 3). Without UG, the (*Generative*) grammar of the respective languages could not be acquired, and without UG, the entire architecture of the *Generative* grammar of a language would immediately be recognisable for what it is, namely highly arbitrary in too many respects. Therefore, without PoS, no UG, and without UG, no Generative Grammar.

That is why PoS tends to be defended tooth and nail. However, as important as UG is for Generative grammar, as unclear has remained its architecture and its modus operandi. What would it mean that a *genetically* coded programme guides children in acquiring the grammar of the language they are exposed to? Nobody knows what the programme exactly is, nobody knows how it connects to cognition and learning, nobody knows why languages are not grammatically uniform, directly reflecting the universal guiding mechanism, and finally, nobody knows why it does not show in the genome.[1] These considerations have led to a wealth of critical literature, see for instance Dąbrowska (2015), Pullum & Scholz (2002), Scholz & Pullum (2002, 2006) and literature cited there. Pullum & Scholz (2002:47) conclude: "*the APS* [= argument from poverty of stimulus]$_{\mathrm{HH}}$ *still awaits even a single good supporting example.*"

In retrospect, this part of Generative theory formation is a typical product of its time. In the early years of the Generative enterprise more than half a century ago, the booming computer science and the discovery of genetically determined *behavioural programmes* in behavioural research,[2] were highly relevant topics. At least at the MIT it must have seemed reasonable to regard the grammar of a language as a genetically determined computation programme. A fitting term presented itself: UG.

In science, people achieve immortal fame for discovering 'laws of nature.' Every educated person is familiar with the name of Johannes Kepler. He earned his Nobel Prize-worthy fame with three laws, viz. the 'universal grammar' of planetary motion. Let's imagine someone positively proves that Homo sapiens are genetically equipped with Universal Grammar. This would be a discovery worthy of a Nobel Prize in physiology. Alas, there will be "no trip to Stockholm" (© Geffrey Pullum) because UG has remained a speculation. Unlike Kepler in *Astronomia nova* and *Harmonices mundi*, no one bothers to exactly define what UG consists of but we are simply told that UG is UG: "*UG consists of the mechanisms specific to FL, arising somehow in the course of evolution of language*." Chomsky (2007: 5). Kepler's hypotheses were empirically confirmed. The UG hypothesis cannot be tested because it remained a constantly changing, vague idea.

Apparently, Generative grammarians not interested in thorough cross-linguistic empirical UG-research anymore. It seems they got stuck in a self-made dilemma. On the one hand, it is now obvious that grammars are not as similar as one NY Yellow Cab is to another. "*Languages appear to be extremely complex, varying radically from one another*." Chomsky (2017:2). Consequently, the system space of UG has to be enlarged. Not all languages are like English. On the other hand, UG is supposed to be the solution of the PoS problem. However, the larger the system space for grammars, the less likely it is that a universal grammar can help children find their way through the maze of a grammar if the universal grammar itself becomes a maze.

Back in the early eighties, the Principles & Parameter-Model promised a potential solution. UG provides parametric options and the grammar learner has to make choices based on the input. Soon there was an inflation of parameters. Later, parametrisation got replaced by equally inflationary, arbitrary sets of features. Parameter valuation reappeared as feature valuation:

[1] FoxP2 is not a *grammar* gene; see Atkinson, al. (2018).
[2] The 1973 *Nobel Prize in Physiology or Medicine* was jointly awarded to Karl von Frisch, Konrad Lorenz and Nikolaas Tinbergen for foundational discoveries in the field of the scientific study of animal behaviour.

> "*I would like to propose the following informal characterization. A parameter is an instruction to perform a certain syntactic action expressed as a feature on an item of the functional lexicon and made operative when the item enters syntax as a head.*" Rizzi (2014: 22).

How children might figure out that they are supposed to identify a feature and how they might determine its value is left to the wondrous creativity of children. In any case, an empirically well-founded *restrictive* theory of parameters or features has never been on the agenda: *"In their totality, these problems suggest that the notion of parametric dependencies runs into empirical problems that should cast doubt on the feasibility of parametric approaches to UG."* (Boeckx & Leivada 2013: 8)

UG is a bold claim that may have "felt" right for a while but couldn't be justified as right. Without empirical confirmation, an unconfirmed claim remains just one of countless unconfirmed claims. This makes it a matter of belief rather than knowledge. It is no coincidence that biologists find the parallel between religion and Generative grammar research somewhat amusing: "*The arguments for intelligent design from irreducible complexity bear an uncomfortable similarity to that originally posited for the necessity of a genetically specified universal grammar.*" (Finlay 2009: 262). The reason for this verdict is as follows:

> "*Bemusement is this biologist's response when straying into cognitive territory, regarding its denizens prospecting for the universals of language and cognition. What could they be looking for, and what would the demonstration of a universal feature of language learning signify to them? If the language prospectors believe the world to be unstructured, the vehicles of perception and production unlimited, the content of communication, and the evolutionary possibilities of the brain relevant to communication unconstrained, then the appearance of "language universals" in independent language learners would be a remarkable and illuminating finding.* [...] *But if any aspect of the world is structured, if available information has predictable content or history, or if information-processing capacities were limited, universals could arise from any or all of these sources, if we may draw parallels with other biological information-transmission devices."* (Finlay 2009: 261)

The PoS argument for UG is based on a claim of *irreducible complexity* from the perspective of learners. According to this assumption, the complexity of grammars cannot be broken down into portions that a child could successfully acquire step by step. In fact, there is a parallel in anthropology, namely the "*birch-tar fallacy*." This and PoS are sub-cases of the *uniqueness fallacy*. The understanding of highly complex capacities as *unique* appears to be deeply entrenched in scholars. The birch-tar episode is a fine illustration.

The production of birch tar – used for hafting tools – by Neanderthals (and homo sapiens) "*was assumed to require a cognitively demanding setup, in which birch bark is heated in anaerobic conditions, a setup whose inherent complexity was thought to require modern levels of cognition and cultural transmission*." Schmidt et al. (2019: 17707). Therefore, the use of birch tar was seen as evidence of modern cognitive abilities and cultural behaviour among Neanderthals in the literature, see Wragg Sykes (2015: 125).

However, the complex method WE would use to produce birch tar today, perhaps with some expert briefing by chemists, is not the Neanderthal way and cannot serve as reference point. Birch tar is more of an unplanned, lucky bycatch than the result of smart process management:

"*We found a previously unknown way to produce birch tar. Instead of creating cognitively demanding structures (underground or in containers), this method consists of simply burning bark close to cobbles in a hearth.*" Schmidt et al. (2019: 17707). Whoever moved cobbles after a birch wood campfire had cooled down could not avoid taking note of the black, extremely sticky substance. The PoS argument is the birch-tar argument of Generativists,[3] as will be argued. Grammars can be acquired without UG. Their (cross-linguistic uniform) properties are properties of their *cognitive evolution* over the past 300.000(!) to 400.000(!) years.[4]

2. What is evidence for an irreducible complexity in grammars?

There is no shortage of claims based on what can only be acquired through innately primed learning of English, that is, guided by UG; see the details in Pullum & Scholz (2002: Section 2.1). So, I shall focus on a *comparative* issue in Section 2.3 of this article, namely on a primary candidate for a UG principle, the *Minimal Link Condition* (formerly known as Superiority).

Over the past twenty years, attention has (been) shifted away from syntactic details and toward more abstract questions of the computational architecture, in particular "merge" and "recursion", which is the topic of Section 2.2:

∴ "*We hypothesize that FLN only includes recursion and is the only uniquely human component of the faculty of language*." [Hauser et al. (2002:1569)].

∴ ["*If these and other cases fall under the same general rubric,*] *then unbounded Merge is not only a genetically determined property of language, but also unique to it.* [Chomsky (2007: 8)].

Let us note that if this were true, PoS would be reduced to a single set, namely merge & recursion, as the specifically human computational capacity. Children would be genetically predisposed to compute recursive linguistic structures without having to laboriously train themselves to do so. So far, so good. But for decades, the debate about PoS revolved around a set of *specific* 'universal' properties of grammars that children supposedly possess without these having been deduced from acquisition data. If these challenges are real challenges for grammar acquisition, then they will not disappear by radically reducing UG. On the contrary. Under these conditions, Generative grammars are no longer learnable for children and are reserved for the self-absorption of a few undaunted syntacticians.

UG or not, there are actually more cross-linguistic grammatical similarities than the mere existence of recursive structures. These were also the subject of UG claims and still require explanation. The following sections are devoted to three of these topics, namely structure dependence (2.1), the resulting recursive structures (2.2), and notorious, allegedly unlearnable restrictions on wh-question constructions (2.3).

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[3] Dan Slobin, cited in Dąbrowska (2015:9), refers to PoS as the "*argument from the poverty of imagination*: *I can't possibly imagine how X could possibly be learned from the input; therefore, it must be innate.*"

[4] Indirect traces of homo sapiens in Britain can be dated back *400.000*(!) *years*, as Davis et. al. (2025) claim, based on fire-making as a uniquely human innovation. Direct fossil evidence of homo sapiens from *300.000*(!) *years* ago comes from North Africa, see Richter et. al. (2017) or Stringer & Galway-Witham (2017).

2.1 Structure dependence

One of the longest-standing candidates for UG status is the *structure dependence* of grammatical regularities and relationships. Even after countless mantra-like repetitions, this argument remains unconvincing, but it became very well-known and people got used to it. Let's start with it therefore. From the beginning, it was a rhetorical masterpiece to present a null hypothesis as a remarkable quality that could only be explained in terms of an innate linguistic quality. In fact, structure dependence is not a unique property of languages, it is a constant of human cognition:

∴ **Memory-encoding**: The brain better encodes and retrieves information when it is organized into a meaningful, *hierarchical structure*. Working memory organisation shows structural sensitivity. We *chunk information* hierarchically.

∴ **Visual processing** and object recognition: The brain organises raw visual stimuli into coherent objects and scenes by processing structural relationships between features. We parse images into objects, objects into parts, and recognise spatial relationships (inside, on top of, behind) that depend *on structural organisation.*

∴ **Problem-solving:** Complex problems are often solved using *hierarchical processing*, where the problem is broken down into *smaller, structured components*

∴ **Reasoning**: Human reasoning often follows *structure-dependent rules.* Analogical reasoning depends on mapping structural relationships between domains**.**

∴ **Music perception** relies on hierarchical relationships such as rhythmic and metric hierarchies and melodic grouping. It organizes individual notes into motifs, motifs into phrases, and phrases into complete melodies. A melody isn't just a sequence of pitches but a *nested structure* where smaller units combine into larger gestures.

It would be surprising if language processing *ignored* consequent hierarchical structuring. The fact that grammars are defined on such structures is by no means remarkable. This is how human cognition works. The examples above demonstrate that across all cognitive modalities, human cognition operates on *hierarchically* structured representations where the relationships between elements matter as much as the elements themselves. The remarkable thing is that it was possible to stylise structure dependence as a *unique* feature of languages and to get many linguists to accept this for a long time.

The following kind of example framed the discussion, first, as showcase of a structure-dependent rule, and secondly, as opportunity for the suggestive question of how children might learn about this fact: How could a child find out that is has to front the structurally *highest* finite auxiliary and not simply the *first* one? How does a child know whether (1b) or (1c) is the grammatically acceptable question to be answered by (1a), as Berwick et al. (2011:1214) ask themselves. In other words, why would a child not trace back the modal to the *nearest* verb in the utterance, which would be “fly”, as in (1c)? In *linear* terms, (1b) would be a “minimal link”, compared to (1c).

(1) a. [Birds that can_1 fly] also can_2 swim
b.*Can_1 [birds that -- fly] also can_2 swim?
c. Can_2 [birds that can_1 fly], also -- swim?

It's structure dependence, it's UG, so is the claim. The first statement is right, but not the second one. The 'test items' crucially contain a relative clause. So, our 'test subjects' will have to be children that are able to master a relative clause. If they do, they interpret "birds that can fly" as the subject of the clause, and they already know that modals follow the subject and precede the verb. The slot that meets this requirement is the pre-VP slot in (1c).

The appropriate answer to the initial question is this: The child already knows that modal verbs follow the subject and precede the verb before they can handle relative clauses. When they have mastered relative clauses, they know that the subject of (1a) is "birds that can fly" and this determines the position of the modal.

It should be clear that the question is ill-posed since the relevant issue is neither a property specific to English nor a property specific to other grammars. The relevant question is this: Why are there no grammars at all with rules and principles that are *not* structure-dependent? In parsing, for instance, 'closeness' matters. This is known as "immediate attachment principle" or "minimal attachment".[5] Here is a famous example. The *closest* potentially finite verb is interpreted as the main verb and this turns (2) into a garden-path construction, with "stop" as the wrongly identified finite verb.

(2) The raft stopped at the jetty sinks.

So why do grammars work in a structure-based way and not by linear arrays? Repeatedly Chomsky emphasises *"the only plausible conclusion, then is that structure-dependence is an innate property of the language faculty,* [...] *ignoring linear distance*" and "*ignoring properties of the externalized signal, even such simple properties as linear order"*. Chomsky (2017: 5).

What should be stressed instead is that structure-based processing is much more efficient than processing information that is coded by linearly-based relations; see Perfors et al. (2011). This is the simple answer. The framing is good rhetorics but bad argumentation. Why would linearity-based properties be "*simple properties"* (see Chomky et al. 2019) and mentally easier to handle than structure-based ones? The simple inversion of a list of items, for instance, is a computation that is very easy to program,[6] very easy to explain, but very hard to carry out spontaneously. No grammar uses list inversion of terminals as a grammatical rule although it is *conceptually* much simpler than structure-based conditions.

Let's return to (1) and ask ourselves what a linearity-based grammar of a wh-clause with auxiliary inversion would look like, assuming that a child has already figured out how to identify the auxiliary verb in the main clause. (3b-d) is what the child faces as the next step. What would be the linear-based rule of Aux-fronting in Wh-questions?

(3) a. Birds that can$_1$ swim can$_2$ dive (deep (into the water))
b. How can$_2$ birds that can$_1$ swim --$_2$ dive?
c.*How can$_2$ deep birds that can$_1$ swim --$_2$ dive?
d.*How deep can$_2$ into the water birds that can$_1$ swim --$_2$ dive?

[5] *Minimal attachment*: When the parser encounters a new word, it attaches it to the current phrase or clause in the simplest way possible, creating the simplest syntactic structure with the fewest nodes. The parser prefers to attach incoming words to the phrase currently being processed rather than closing it off and starting a new one.

[6] In Prolog, a simple command like "reverse ([a,b,c,d,e,f,g], Results)" yields the inverted list [g,f,e,d,c,b,a]. Our brains evidently do not provide a list-reverse function.

It is up to the reader to – *first* – imagine a grammatical rule that correctly covers auxiliary inversion *without* resorting to terms denoting something like '*constituent', 'phrase', 'what belongs together'* or '*what is a unit*' and – *second* – to show that it is 'simpler' than the structure-based one, namely, that the interrogative *phrase* goes to the clause-initial position and the finite auxiliary follows it. "*Ignoring linear distance"* is no UG-effect. It is the result of resorting to the only efficient way of information processing for a human brain. Grammars based only on linearity are unsuitable for our mental computing capacities.

Our brain works highly efficient when it applies chunking[7] (cf. Miller 1956, Gobet & Lane 2012, Norris & Kalm 2021) but is much less efficient when it has to apply operations on unstructured, only linearly organized bits and pieces. This is a consequence of (cognitive) evolution and evolution is a competitive process that rewards efficient methods of information processing.

Structure-dependence is the linguistic instantiation of a more general property of human information processing which is the solution to a general bottle-neck problem in the flow of information. The bottle-neck is the limitation of our working memory. For PCs, there are RAM extensions; human brains, however, are born with their memory capacity, without later memory-expansion options (via bio-chips?). Therefore, an increase in the amount of information to be processed can only be achieved by structuring it into portions that can be processed more quickly. This is known as "chunking".[8]

Structure dependence is one side of the coin, with chunking as the other side. Structure dependence is the linguistic consequence of chunking as an efficient method for recoding information. Syntactic constituents are labelled chunks. Moreover, chunking is a domain-general capacity that is operative in various cognitive modalities, not excluding language processing. It should therefore come as no surprise that grammatical rules tend to function via syntactically labelled units (also known as phrases) rather than serial arrangements. Structure-dependency is another way of describing the fact that rules of grammar operate with structured units and not on serial representations of terminals.[9] This is neither surprising nor does it justify the invoking of an innate UG. From the perspective of the general functioning of our cognitive abilities, structure dependence is the null hypothesis and a constant in human cognition.

Let's briefly reverse the perspective. How clumsy would symbol processing work without structure dependence? Here is an example from the vicinity of language processing, namely the grammar of numbers. Imagine you were a Roman during Julius Caesar's time and had to square the following number: CMXCVII. Ask an AI system how to proceed and you will be presented an abacus-based calculation that covers two pages. In our decimal structure-dependent 'grammar of numbers', the Roman number is written 997, and nearly everyone[10] can square such numbers without any tools. This is the advantage of structure dependence in its purest form.

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[7] Chunking involves recoding the input into a more efficient code in the same 'vocabulary' (see Norris & Kalm 2021:9).

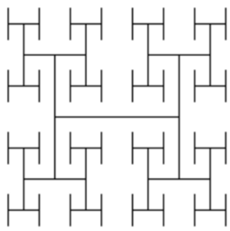
[8] The following paragraphs paraphrase the discussion of the question, which is elaborated in Haider (2025a).

[9] According to Dirlam's (1972) mathematical analysis of efficiency, the "*most efficient chunk size*" is 3 to 4 items per chunk. Van Benthem et al. (2016: 85): *"In natural language, ternary predicates are enough to express the most complex verb pattern you can get, but logical languages can handle any number of arguments*. Voilà.

[10] 997+3 = 1000. 997-3 = 994. 994x1000 = 994000. So: 997^2 = 994000+3^2 = 994009. Why? $a^2 = (a+b)*(a-b)+b^2$.

2.2 Recursion & merging

Recursion has been hyped as a "cool" topic, but the discussion about it is uncool. Recursion is stylised into a uniquely linguistic ability that supposedly forms the central element of grammar processing. As for recursion, "*we hypothesize that FLN only includes recursion and is the only uniquely human component of the faculty of language*", and, as already quoted above: "*We suggest that FLN – the computational mechanism of recursion – is recently evolved and unique to our species*" (Hauser et. al. 2002:1573). Please note the ambiguous term "*computational mechanism.*" It is ambiguous between a narrow and a broad reading. The narrow reading is recursion in *algebraic* computation. In this sense "NP → N° (NP)" is a recursive rule and the programme that contains this rule is the "*computational mechanism.*" In the broad reading, "the computational mechanism of recursion" is our cognitive capacity for language processing. Surely, the capability of processing *recursive structures* is not an exclusively linguistic capacity. Here is an example:



How often do the initials auf the author – HH – *occur in the graphic*? If you try to count them, you will unavoidably note that *pairs* of H come in two sizes: 8 pairs in small font and two big pairs, and a single big H as centre.

Figure 1: Graphic recursion of HH

Generative grammar assumes – without any psycho-linguistic confirmation – that language processing is an *algebraic* skill. Elements must be combined into larger units by "merge" and transformed according to rules. This is far from plausible. The human brain is bad in algebra but excellent in pattern processing. Language processing is a pattern-based skill. Grammars define structures. Processing involves the mapping between one-dimensional arrays (= phonological form) and at least two-dimensional, labelled box-in-box structures. Reception of speech is 1-D (linear array of phonological form) to 2D mapping (= syntactic structure); production is the inverse, namely 2D-to-1D mapping. This is where our brain excels. If the mapping succeeds, the utterance is structured grammatically. Chunking is an essential technique for mapping. Recursion is but another way of describing *chunks that contain chunks* of the same category. Since chunks may consists of chunks, recursion of chunks is an expected consequence for sufficiently powerful information processing systems. According to Perfors (2010:159), the optimal grammar is a depth-limited recursive grammar.

The supposedly innate "merge" operation is chunking turned upside-down. Chunking works in such a way that larger units are *perceived* (rather than *generated*) as complex units made up of atomic units, piece by piece. Structure processing is not reverse engineering based on a structure-generating merge operation. Structure processing is *pattern detection*, not *structure generation*. "Merge" is but the operational mimicking of pattern detection. However, structures are not mentally *generated*. They are *projected* over strings in language in-take. Consequently, in language out-put, *speakers* arrange *strings* in such a way that they can project well-formed structures on them. "Merge" is a highly misleading operational detour on the way to structure projection.

It is truly amazing that the fact that structure-dependence and recursion are intertwined seems to have received little attention. In fact, these are two sides of the same coin, and the name of the cognitive currency is "chunking." Incidentally, George Miller published his ground-breaking paper on chunking at the time when "Syntactic Structures" went to press.

In the meantime, it has become apparent that Hauser, Chomsky and Fitch's bold claim that the mastery of recursive structures in communication is a unique feature of Homo sapiens was too daring. Biologists object: "*There is more empirical evidence of recursion in animal communication.*" (Barón Birchenall 2016: 22). Currently, evidence is available showing that other species master tasks that involve recursion:

"*We reveal that crows have recursive capacities; they perform on par with children and even outperform macaques. [...] These results demonstrate that recursive capabilities are not limited to the primate genealogy and may have occurred separately from or before human symbolic competence in different animal taxa*." Liao et al. (2022: 1). Comparisons between humans and other primates shows that "*the capacity itself is not unique to humans.*" Ferrigno et al. (2020: 1). Lameira et al. (2024) present immediate evidence for recursion in *vocal communication* from the analysis of wild orangutans:

> *„Findings represent a case of temporally recursive hominid vocal combinatorics in the absence of syntax, semantics, phonology, or music. Second-order combinatorics, 'sequences within sequences', involving hierarchically organized and cyclically structured vocal sounds in ancient hominids may have preluded the evolution of recursion in modern language-able humans."* Lameira et al. (2024:1).

In this context, the linguistic controversies surrounding the Pirahã language seem somewhat academic. Even if it may be true that everyday Pirahã communication functions well without *subordinate* clauses, that would per se not contradict the claim that recursion requires UG. The absence of recursive structures in Pirahã, if true, would only refute the claim that UG-based grammars *must* display recursive structures. However, it is possible to not exploit the full grammatical potential.

"**Merge**" is the other candidate said to be innate: *"Merge, and the condition that it can apply without bound, fall within UG"* Chomsky (2007:5). Seriously?! Are we told that "*Me Tarzan, you Jane*" could not have been produced and understood without being innately gifted with "merge"? What would be the alternative? Is there anyone who would dispute that we generally understand the properties of complex structures as a function of the properties of its simpler, individual parts and the specific ways in which those parts are combined? Why would grammar be different? It isn't.

The 'HH-structure' in Figure 1 is compositional. One way to describe this is by "merge": Start with an atomic H, which itself is a graph with four 'ends'. Next, merge each end with an atomic H, attaching it at its centre. Then you repeat this merge-operation for each one of the resulting open ends. The result is Figure 1. This, however, is just *one* possible description of the H-graph, namely the *operational* one. It reconstructs the production process. *Merge* is an operational description. In computational terms, it is part of a generative programme.

But, there is an equivalent characterisation without "merge." When I process Figure 1, I see a diagram that I did not draw, but which I can still capture without redrawing it. I understand its structure by the spontaneous insight that it consists of an H, with an H attached to each of its four legs, which in turn have an H on each of their legs. This is the *constitutional* characterisation. This is a Gestalt description as a result of the analytical processes of my brain. I am not 're-merging' it, but rather perceiving its Gestalt property.

Generative grammar insists that our grammar capacities have to be modelled *operationally*. It would be very surprising if it were so. Compelling evidence is missing. What Generative Grammar demands of us is as follows. We have to believe that a sentence is processed by reconstructing it. Our grammatical competence, we are told, consists of understanding the step-by-step construction of any sentence, relying on *merge* and *move*. If we succeed we have understood the sentence and feel somehow that it is well-formed. Grammatical competence is presented as a theorem-proving competence:

> "[...] *generation of expressions is similar to other* (!)$_{\mathrm{HH}}$ *recursive processes such as construction of formal proofs. Intuitively, the proof "begins" with axioms and each line is added to earlier lines by rules of inference or additional axioms.* (Chomsky 2007: 6).

Human theorem proving in other domains lacks the *instantaneous* quality of grammaticality judgements and it completely disregards grammar acquisition. Generativists do not bother demonstrating that the cognitive capacities of little children would be sufficient for acquiring the grammatical axioms needed to activate their innate proof-generating talents. In the absence of any psycho-linguistic evidence, we are fobbed off with speculations. At least during a lifetime, we will not be able to understand the true operational side of neurocognitive grammar processing, but we may be able to understand the *constitutive* quality of the task that our brain accomplishes. Grammatical competence is not 'operational', it is 'constitutional.' Grammatical structures are *recognised* online, but they are not *generated* online.

2.3 In-situ wh-subjects as a concrete test case

Let us finally examine an exemplary case as it has to be examined if one wants to use it as argument for or against PoS: What are the facts? What is the grammatical regularity? How can it – or can't it – be acquired bottom-up?

The facts: There seems to be a general restriction on the interpretation of in-situ wh-items in languages in which a *single* interrogative wh-item is fronted, such as in English or German and other languages. This restriction, in addition, interacts with a clause-structure difference, as will turn out. In the examples (4), "*what*" is in situ and therefore a wh-item that is "dependent" on a fronted one. It cannot be fronted itself because there is already a wh-item in the target position. The interpretation of the in-situ wh-item is linked to the licensing wh-phrase.[11] So, (4a) is interpreted as a double question that is answered by pairs of places and objects. (4b) is an indirect question with paired wh-items that specifies what it is that we "need to know", namely pairs of places and objects. (4c) is ambiguous. "What" could be linked either to "who" or to "where." (4c) is either a double wh-question or a simple wh-question that contains a 'doubled' indirect question.

(4) a. *Where* did he hide *what*?
b. We need to know *where* he has hidden *what*.

[11] It receives the same interpretation as the wh-item it depends on. The dependent items in (i.) an (ii.) are "whom", "what" and "when". In each case, they are interpreted like "who", that is, as a matrix-question item in (i) and as an indirect-question item in (ii): i. Who told *whom what*, and *when*?
ii. It is not known who told *whom what*, and *when*?

c. *Who* knows where he has hidden *what*?

The examples in (5) illustrate an iconic contrast for Generative Grammar in the 1970ies, which Chomsky (1973) named the "Superiority" constraint. Later, it was re-named "Minimal Link" condition. These names are misleading because it soon turned out that the contrast is independent of movement and not universal.

(5) a.*Where did *who* hide something.
b.*We need to know where *who* hid something.
c.*Who knows where *who* has hidden something?

The restriction only holds for English and languages similar to English, but not in SOV languages, such as Japanese or German (6); see Lasnik & Saito (1984), Haider (1986: 114). Minimal distance cannot be a universal, grammatically causal factor.

(6) a. Wo hat wer etwas versteckt? (= 5a)
b. Wir müssen wissen, wo *wer* etwas versteckt hat. (= 5b)
c. Wer weiß, wo *wer* etwas versteckt hat? (= 5c)

Chomsky himself (1981:236) presented a cue for the identification of the relevant grammatical factor that rules out structures as exemplified in (5) but he did not pursue it. The deviance in (5) is not due to any minimal-distance requirement. It is ruled out by the same condition that rules out the examples in (7). This is not about "superiority" or "minimal distance" since there is no crossing movement involved.[12]

(7) a. I know perfectly well who thinks (that) she/**who* is in love with him.
b. I don't know who would be happy that she/**who* won the prize.
c. I don't remember who believes that she/**who* read the book. Chomsky (1981:236)

The German equivalents of (7a-c) are acceptable (8a), as expected, since in SOV languages, the subject position is VP-internal. What (7) shows is that (5a) to (5c) are unacceptable for the very same reason, namely a wh-subject that is in-situ and dependent in its interpretation on a preceding wh-item.

In English, unlike in German or Japanese, subjects are obligatorily assigned to a Spec-position. This explains the difference, and there is *independent* evidence that in German, wh-elements that are in a Spec position in which the only[13] interpretation is interrogative cannot be interpreted as dependent either. In (8b), the wh-item in the spec of the embedded V2-clause cannot be interpreted as a dependent interrogative. The clause-initial spec-position is the position for interrogative interpretation of a wh-pronoun in German, in main as well as in embedded clauses.

(8) a. *Wer* behauptet, [*dass* [sich hier *wer* versteckt haben könnte?]]
who claims that REFL here who/someone hidden have could
b. *Wer* behauptet, [[er/**wer*]$_{\text{Spec-C}}$ [könnte [sich hier verstecken?]]]
who claims he/*wer/*someone could REFL here hide

In German, many verbs (verba declarandi or putandi) can either take a finite sentential

[12] "*Covert movement*!" would not be a viable excuse since in-situ wh-pronouns can occur within robustly opaque domains that would block any movement. There is no such 'movement.' See Haider (2024: Sect. 2.3)

[13] In their base position, wh-pronouns such as *wer* (who), *was* (what), *wo* (where) can be interpreted as indefinite pronouns.

complement in the form of a C°-introduced clause, where C° = *dass* (8a), or a V2 clause (8b). In this case, the dependent wh-element of the complement clause must not end up in the spec position of the V2-complement. Even the indefinite-pronoun reading is inaccessible for "wer." In Spec, a wh-pronoun must be interpreted as an *independent* wh-pronoun. This rules out the wh-pronoun of a double question in (8b) since it would have to be dependent on the matrix wh-item.

Before we ask ourselves now what role UG plays in such a constellation, we need to examine an undisputable instance of "Superiority". The contrast in (9) does not involve the subject. In (9b), the base position of the fronted wh-item is in the infinitival object clause. It is fronted across a wh-in-situ item in the matrix clause, in violation of the minimal-link economy.

(9) a. Who did you persuade -- [to buy *what*]?
b.*What did you persuade *who* [to buy --]?

However, German again contrasts with English with respect to (9b). In German, both equivalents of (9), namely (10a,b), are acceptable because the pronouns are *grammatically* distinct from each other. (10c) is unacceptable. Here, the two occurrences of "*wen*" are in conflict. There is no conflict in (10d). The case forms are distinct.

(10) a. Wen hast du überredet --, [dort *was* zu kaufen]? (= 9a)
b. Was hast du *wen* überredet, [dort --zu kaufen]? (= 9b)
c.*Wen hast du *wen* überredet, [dort -- zu besuchen]?
whom did you *whom* persuade [there to visit --]
d. *Wen* hast du *wem* versprochen, [dort -- zu besuchen]?
whom$_{\mathbf{Acc}}$ did you (to) whom$_{\mathbf{Dat}}$ promise [there to visit --]

What we see here is evidently a processing-based restriction. When the parse of (10c) reaches the second "wen", the buffer already contains a wh-item with the *very same grammatical form* and is blocked thereby. Now we can disentangle the crossing restriction of (9b) from the wh-in-Spec restriction of (5) and (7), with the help of a comparative perspective: In English, the wh-items in (9) cannot be distinguished by their grammatical form (only by content), and the structure becomes deviant. In German, the same restriction applies to (10c), but not to the structurally equivalent (10b)[14] or (10d).

Let us now turn to the identification of the *grammatical regularity* behind all these data. The relevant generalisation – see Haider (2010, Chapter 3.4) – is as follows and it is intentionally formulated in terms of *epistemological priority* (= primary quality of the data for a learner):

(11) In *multiple*-wh constructions, a wh-element in the position of a sentence where it appears as an interrogative pronoun in a *single*-wh question is interpreted in the same way as in single-wh questions (i.e. not as dependent).[15]

The SOV-SVO contrast thus reduces to a contrast in clause-structuring and the morpho-syntactic inventory. In English (and languages similar to it), it singles out the subject. The interroga-

[14] "Was" (what) and "wen (whom) are formally different (not only semantically). "Wen" is uniquely accusative, "was" is a form that can be nominative or accusative. So, "wen" and "was" are *formally* different.

[15] Subjects of ECM-infinitival complements are grammatical in situ although "who" is in the Spec-position of the infinitival functional head *"to"* the reason being, that this is not a position of interrogative interpretation: i. Who expects [*who* to prevail]? ii. It is unclear who would expect [*who* to prevail]?

tive position for non-subjects, precedes the subject position, but a wh-subject in its pre-VP position is not fronted additionally because it is in a 'fronted' position already, that is, in the spec-position of a functional head. In SOV (and VSO), the surface position of the subject is identical with the VP-internal argument position, that is, its argumental base position. There is a VP-external, mandatory functional subject position only in the [S[VO]] clause-structure. In sum, a wh-subject is interpreted as interrogative in its subject position in English. Hence it cannot receive a dependent interpretation.

How can the regularity be acquired? Here we are finally at the issue of PoS. Die-hard Generativists need to update their outdated UG version. The decisive factor is not a minimality condition on wh-chain formation – which would amount to a parsing-based economy restriction – but rather a structural property, i.e. a restriction on the interpretation of wh-elements in their actual positions.

It is an intriguing quality of this data area that the relevant regularity is entangled with clause structure. It is not an independent property of wh-movement constructions that applies across languages, as would have been the case for "Superiority." It appears to be complex since it involves a condition that depends on a cross-linguistically variant quality, namely the structural quality of subject positions in simple questions compared to multiple questions. In reality, the task is simple: If the wh-subject is in a position for an interrogative interpretation in a simple clause, it *cannot* be interpreted as a dependent wh-item, otherwise it can. The task for the learner is a straightforward pattern matching task.

English learners must be able to pay attention to the *structural* difference between subjects and objects, which does not play a role in German, Japanese, Russian, etc. In English, the pragmatic equivalent of (12a), with "who(m)" as the sorting key, can be achieved only by paraphrasing as in (12b) because the immediate equivalent (12c) is grammatically not admissible.

(12) a. Wen beeindruckt **was** am meisten?
whom impresses what as-the most?
b. Who is most impressed by **what**?
c.*Whom does **what** impress most?

How can learners of English – if the wh-subject is in its Spec position – or German – if a wh-element is in Spec of an embedded V2 sentence – acquire these distinctions when they are hardly ever confronted with such data (except for syntax classes, which they don't attend)? Let us assume – without having verified this typologically – that (11) holds universally and thus excludes dependently interpreted in-situ Wh-subjects in every [S[VO]] language that fronts only a single wh-pronoun. Does this qualify a technical version of (11) as a principle of universal grammar? After all, the poverty of stimulus in relation to such constructions can hardly be undercut.

How could we be sure that the restriction is (un)learnable or not? We can only observe that native speakers of English behave accordingly. What is triggering their behaviour is inaccessible to the observer. It must a consequence of how the system is acquired. Could a grammar, shaped by cognitive evolution, steer acquisition in this direction? It seems that this is what happens.

Here is a sketch: Grammar acquisition proceeds from simple patterns to more complex ones. The simple pattern is the key for the decoding of more complex ones. In a nom-acc language, the simplest pattern of a wh-construction is a direct question with a wh-subject. In English, there is no change of place involved and no *do*-support. This is true for indirect questions as well. The learner easily finds out that in such a case, the place of "*who*" is the very same place in which the subject occurs in the answer. In the majority of wh-questions, the answer-item typically occurs clause-internally which is also a cue that the corresponding wh-element is fronted. This simple constellation is sufficient to single out the subject as the only wh-element that is interpreted in its 'answer-position'. So, the child is driven to conclude that for subjects, the question position is the answer position and then proceeds to the stronger, bi-conditional version, namely, that the answer position is also the question position for the subject.

The weak version is conditional, the strong version bi-conditional: A wh-subject is in the answer position, *and* the answer position is the position in which it triggers a question interpretation. Young children tend to spontaneously interpret standard conditionals ("If P, then Q") as bi-conditionals ("If and only if P, then Q") and weaken it later, if necessary. This is a phenomenon often observed in cognitive-development research. The result of strengthening is (11). Without 'reinforcement,' we would expect a learner not to regard the non-occurrence of a pattern as proof that it is excluded.

The wh-subject in Spec initiates a question, which rules out a dependent interpretation. The same reasoning applies to German, but in the reverse direction. In German, a V2-language, a Wh-item in the Spec-position of a V2-clause has only a question-reading (13b), while in the base position it can be interpreted as an indefinite pronoun (13a). (13c) is ambiguous.

(13) a. Heute hat *wer* angerufen
today has *someone* called
b. *Wer* hat heute angerufen
who/*someone has today called
c. Wie oft hat heute *wer* angerufen?
How often has today *who/someone* called

A learner never meets a wh-item in the clause-initial Spec-position that is *not* used as an interrogative expression, including rhetoric questions. This is enough evidence for safely inferring that a wh-item in the Spec of an embedded clause cannot be interpreted as dependent element.

In summary, it should be noted that with regard to the learnability of wh-positioning in German or English, there is no need to declare a state of emergency that would justify invoking UG as last resort measure. The German-English contrast with wh-subjects follows from a contrast in sentence structure and is representative for SOV-languages and [S[VO]]-languages, respectively. The residue of Superiority – No crossing dependency if there is a grammatically non-distinct intervening item – is a processing constraint. German has four morphologically marked cases, English has none. So, there are more contexts with blocking interveners in English.

3. No more UG and PoS – There is a better option

The questions that led to PoS and UG were good *questions*, but the *answers* were not good. Nevertheless, for a long time these were the only theory-backed answers to these questions. Scholars are conservative. They would rather live with an awkward theory than have no theory (aka coherent narrative) at all. So, they would not abandon a familiar framework simply because it is empirically unsatisfactory. They would rather try to patch it up. Theories are not discarded by refutation; they are abandoned in favour of a better theory, which was not yet in sight.

Proponents of Universal Grammar (UG) claim that the brain of Homo sapiens – and surprisingly, only of Homo sapiens – has adapted to language processing within a minimal period of time for such an evolutionary leap, and that this has resulted in a genetically determined ability to acquire and process grammar.[16] As the past decades show, this was not the highway to success. According to an apocryphal insight, we should acknowledge that we cannot solve our problems with the same thinking we used when we created them. All we have to do is to reverse the PoS argument and ‘put the horses BEFORE the cart.'

Grammars are learnable *not* because *brains* are pre-programmed for them, but because *grammars* have been programmed for brains through constant selection in the course of cognitive evolution in the past 300.000 years; cf. Haider (2021). Supposed UG principles, i.e. cross-linguistic grammatical invariants, are results of *convergent cognitive evolution* of grammars. The supposedly ‘innate knowledge’ that children are said to possess for language learning is not necessary at all. Millennia of cognitive evolution have led to grammars tailored to language acquisition. Through constant selection, they have developed in such a way that they can be acquired using the available learning mechanisms. A grammar variant that is more difficult to acquire than another variant could not have spread. It would have been sieved out.

The Baldwin effect in evolution – see Haider (2025b: Sect. 4.9) – provides a mechanism for how grammatical systems can become progressively better at exploiting the cognitive resources of their hosts, leading to the seemingly miraculous fit between human cognitive capacities and grammatical complexity. It suggests that much of what looks like innate linguistic knowledge is actually the accumulated result of countless generations of grammars having evolved to be learnable with the most advanced primates’ cognitive abilities.

These few paragraphs are intended as an outline of a theory which has already been elaborated in several articles in the last ten years. They are easily accessible and downloadable for anyone interested.[17] Overall, the predictions from the UG perspective are contrary to the cognitive-evolution perspective and most likely wrong. If UG were the task master of human grammars, grammars should *converge* on default values. What we see is ongoing diversity, cf. Hammarström (2016), and this is predicted by evolution, and in particular, by cognitive evolution.

In this perspective, there is no reasonable justification for PoS. Unlearnable grammar variants could not spread simple because they could not replicate without brains that acquire and use them. The grammars of human languages have developed in a sequence of cycles of language acquisition in childhood and subsequent use. This is working since more than 300,000 years, with the outcome of each cycle forming the basis for the next acquisition cycle. During these millennia, the grammars had time enough to get adapted to the demands of human language processing capacities, recruited from domain-general information processing, which streamlined grammars by cognitive evolution.

[16] Speakers of English seem to be favoured, as we are told that English grammar perfectly matches UG.
[17] https://ling.auf.net/lingbuzz/_search?q=Haider

We finally should humbly bear in mind that our – highly limited – knowledge of the history of grammars is based on available text corpora that cover a period of at most a few thousand years. This accounts for less than 1% of the time since Homo sapiens, a species known to be loquacious, has come into being. This is time enough for the adaptive *cognitive* evolution of grammars, but not enough for a *genetic* evolution of UG. Unfortunately, we have no access to it. Grammars do not fossilise.

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Appendix %

Appendix

∴ Corpus data on wh-subjects in situ

Projekt Gutenberg
https://www.projekt-gutenberg.org › keltisch › keltisch

Keltische Knochen

Wo hat wer gesessen? Wann hat wer hier gesessen?«

Kosmos Theater
https://kosmostheater.at › Produktionen

KORALLI KORALLO

Vielleicht so: „Wir wissen **nicht wo wer** aufhört, wo wer anfängt."

WDR
https://www1.wdr.de › Stichtag

2. Februar 1995: Spiel "Die Siedler von Catan" erscheint

02.02.2020 — Hier überlegt man die ganze Zeit mit: **Was hat wer** bekommen?

historia-universalis.fm
https://historia-universalis.fm › ...

HU200 – Eine kleine Menschheitsgeschichte [Part V]

06.03.2022 — **Wen hat was** überhaupt interessiert? Im tagesaktuelen Kontext

∴ Data rating by AI-system

(1) Are the following sentences grammatically correct German?

a. Wen interessiert, dass uns was gestört hat?
b. Wem fiel auf, dass uns was geärgert hat?
c. Wo steht geschrieben, dass uns was erwartet?
d. Wie kann man herausfinden, dass hier was fehlt?

(2) Are the following English translations of these sentences grammatically correct English?

a. Who cares that what bothered us?
b. Who noticed that what annoyed us?
c. Where is it written that what awaits us?
d. How can we find out that what is missing here?

Answer of the AI-system (ChatGPT):

„Short answer: **Yes**, all four German sentences *can* be interpreted as double questions, i.e. questions containing two interrogative elements (*Wh-Wh-questions*) whose answers naturally come in pairs. They are grammatically correct as such."

Summary

Sentence	German version	English translation
a	✓ correct	✗ incorrect
b	✓ correct	✗ incorrect
c	✓ correct	✗ incorrect
d	✓ correct	✗ incorrect